

UNIT 4: Microprocessors And Microcontroller .

Microprocessors:

Defⁿ:

- A programmable device that takes in numbers, performs on them arithmetic or Logical operations according to the program stored in memory and then produces other numbers as a result.

1. 8085 Microprocessors:

Introduction:

- It is a 40 pin I.C. (Integrated Circuit)
- It uses a +5V D.C supply for its operation.
- It is an 8-bit microprocessor
- Its speed of operation is about 3/5/6 MHz
- It has 16 address lines i.e $2^{16} = 64 \text{ KB}$ memory.

Fig. 5.3, shows the block diagram of Intel 8085 microprocessor.

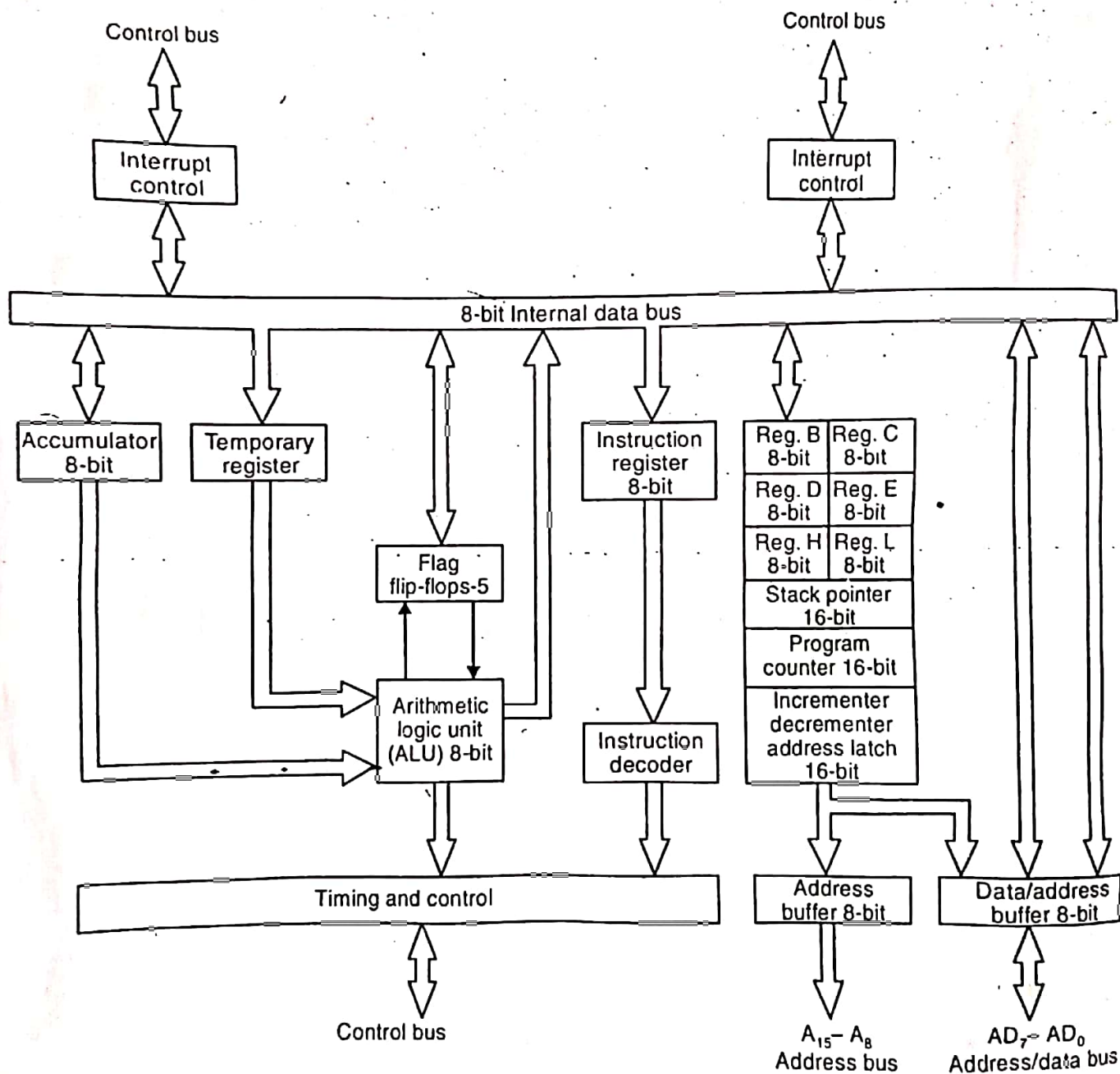
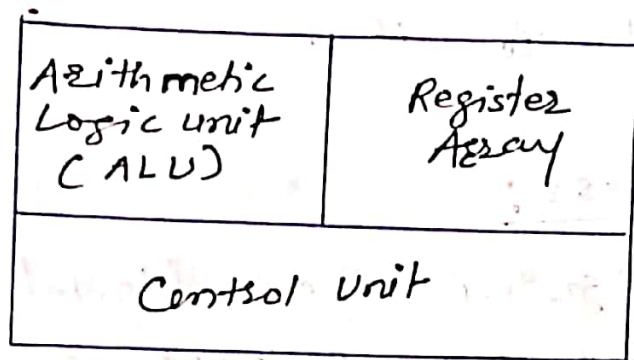


Fig. 5.3. Block diagram of Intel 8085 microprocessor.

→ The microprocessor consists of the following three segments as shown in figure below:



1. Arithmetic and Logic unit (ALU):

- The ALU performs arithmetic operations such as addition and subtraction, logic operations such as AND, OR and exclusive OR. Results are stored either in registers or in memory or sent to output devices.

2. Timing and Control unit:

- This unit is a section of CPU.
- It generates timing and control signals which are necessary for the execution of instructions.
- It controls data flow between CPU and peripherals (including memory).
- It provides status, control and timing signals which are required for the operation of memory & i/p/o/p devices.

- It Controls the entire operations of microprocessors and peripherals connected to it.

3. Registers:

- Registers are digital devices used by the microprocessors for temporary storage and manipulation of data and instructions.

Intel 8085 microprocessors has the following registers:

i) Accumulator (Acc):

- It is 8-bit special purpose register. It is also identified as Register A.
- All logical and arithmetic operations are performed in Accumulator.
- It also helps to store the result of the operation.
- The accumulator may store the operand, in arithmetic and logic operations.

ii) General purpose registers:

- In 8085 microprocessor there are six 8-bit general purpose registers. These are identified as B, C, D, E, H and L as shown in figure.
- The registers can work in pairs to hold 16-bit data & their pairing combinations are B-C, D-E and H-L.
- These registers only be used to store temporary results of operations hence called as temporary registers.

iii) Stack pointer (SP):

- It is a 16-bit special function register.
- The stack is a sequence of memory locations set aside by a programmer to store/retrieve the contents of accumulator, flags, program counter and general purpose registers during the execution of a program. Any portion of the memory can be used as a stack.
- In this register, data is stored temporarily on first come and last go basis.

- The stack pointer is incremented when data is push. (data inserted)
- The stack pointer is decremented when data is pop. (data removed)

iv) Program Counter (PC):

- It is a 16-bit special-purpose register and is used to hold the memory address of the next instruction to be executed.
- It always contains the 16-bit address of memory location where next-executable code is stored.
- The microprocessor uses the program counter (PC) for sequencing the execution of instructions.

v) Instruction register :

- Instruction register is a 8-bit register
- It holds the instruction currently being executed or decoded.
- The instruction register holds the operation code or instruction code of the instruction which is being decoded and executed.

vi) Temporary register :

- It is an 8-bit register associated with ALU.
- It holds data during an arithmetic / Logical operation.
- It is used by the microprocessor and is not accessible to programmer.

vii) Flag Register :

- In 8085 microprocessor, flag register consists of 8-bits and only 5 of them are useful (active).
- It is a special purpose register.
- Depending upon the value of result after any arithmetic and logical operation the flag bits become set (1) or reset (0).

viii) Instruction decoder :

- Data from instruction register is sent to the instruction decoder, where microprocessor decodes it and then translates into specific actions.

ix) Data and address bus :

- The data bus of intel 8085 microprocessor is 8-bit wide and hence 8 bits of data can be transmitted in parallel form to the microprocessor.
- But memory addresses are of 16-bits, so this microprocessor requires a 16-bit wide address bus.
- The 8 most significant bits of the address are transmitted by the address bus i.e. A-bus (pins A₈ to A₁₅).
- The 8 least significant bits of the address are transmitted by address/data bus, i.e. AD-bus (pins AD₀ to AD₇).

Applications :

- Washing machines,
- Microwave Ovens,
- Mobile phones.
- Traffic light control
- Speed control of motors
- Security systems.... etc.

2. Programmable Logic Controller : (PLC)

⇒ PLC is a digital electronic device that uses a programmable memory to store instructions and to implement functions such as logic sequencing, timing, counting and arithmetic in order to control machines and processes.

A PLC consists of the following main components :

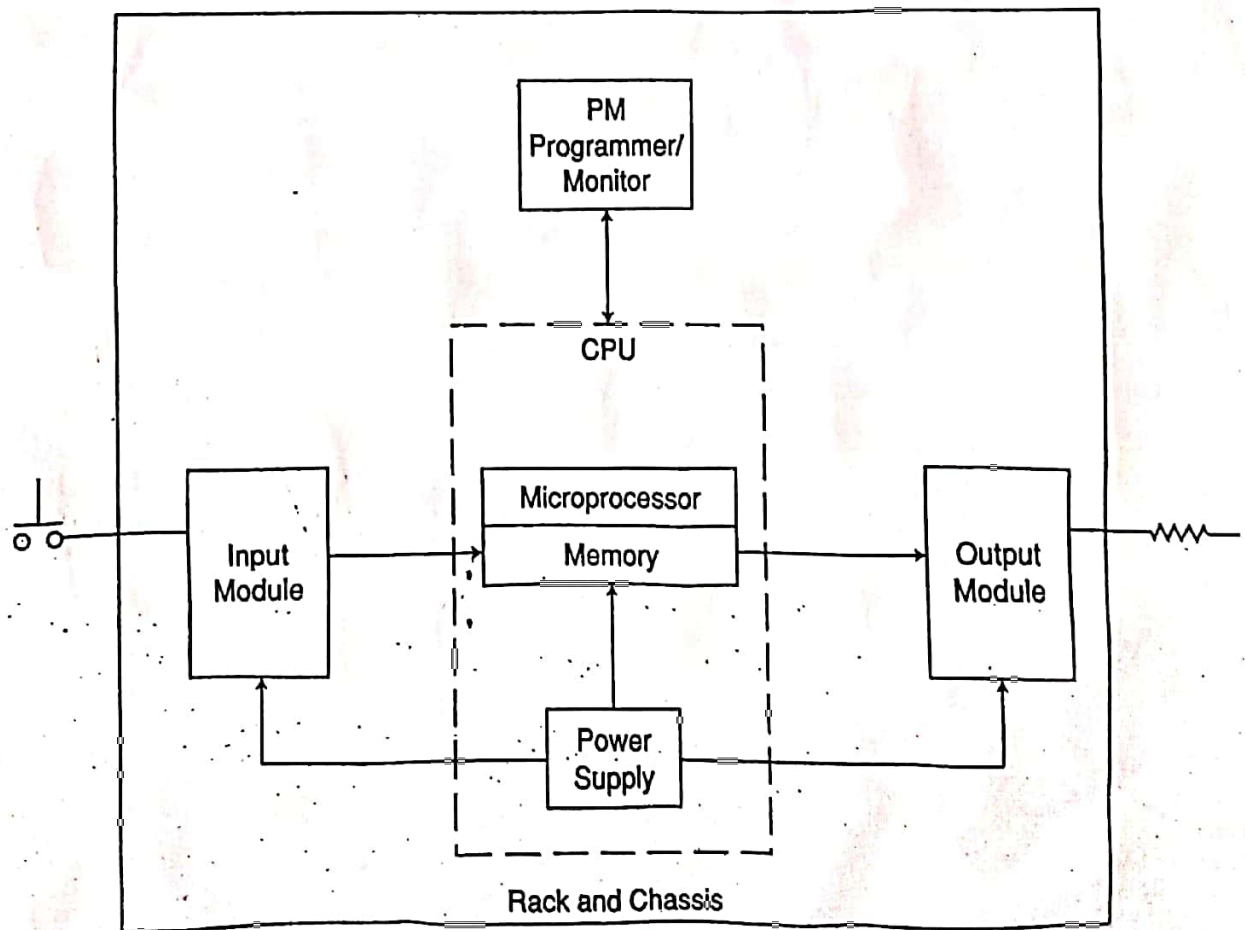


FIGURE 1-4
PLC System Layout and Connection

1. Central processing unit (CPU).
2. Programmer / Monitor
3. I/O Modules.
4. Racks and chassis.

1. Central processing unit (CPU) :

- It controls and processes all the operations within the PLC. It is brain of the system, which has subparts:

- a) Microprocessor : The Computer center that carries out mathematic and logic operations.
- b) Memory : The area of CPU in which data and information is stored and retrieved. Holds the system software and user programs.
- c) Power supply : The electrical supply that converts alternating current (AC) line voltage to various operational DC values. In the process, the power supply filters and regulates the DC voltages to ensure proper computer operation.

2. Programmer / Monitor :

- The programmer / monitor (PM) is a device used to communicate with the circuits of the PLC.
- Hand-held terminals, industrial terminals and the personal computer exist as PM devices.
- In hand-held unit, input takes place through a membrane keypad and the display is usually a liquid-crystal display (LCD).
- With the industrial terminal or personal computer, more complex, typewriter-type keyboards and cathode ray tubes (CRTs) are employed.

3. I/O Modules :

- The input modules have terminals into which outside process electrical signals, generated by sensors or transducers, are entered.
- The output modules have terminals to which output signals are sent to activate relays, solenoids, various solid-state switching devices, motors and displays.

OR

— The input modules receive signals from switches or sensors and send them to the processor and the output modules take back the processor signals to the control devices like relays or motor starters.

+ Racks and chassis :

— Racks and chassis is a hardware assembly that helps in housing the processor, communication and input-output modules. i.e. the PLC parts are mounted on racks and chassis.

— It works in the following ways :

a) It distributes the power evenly.

b) It contains the different i/p & o/p modules.

c) It acts as a communication link between I/O modules and the CPU.

*.

Advantages of PLC :

- 1> Very fast
- 2> Easy to change Logic i.e flexibility.
- 3> Low power Consumption.
- 4> Reliable due to absence of moving parts.
- 5> Easy maintenance due to modular assembly.
- 6> Capable of handling of very complicated Logic operations.

*.

Applications of PLC :

- 1> In traffic light signal system.
- 2> In Nano technology and robotics.
- 3> In automatic transfer switches.
- 4> In industrial manufacturing.
- 5> In medical instruments and technologies.
- 6> In machining, packaging, material handling, automated assembly.
- 7> In washing machines etc.

x.

Selection of a PLC :

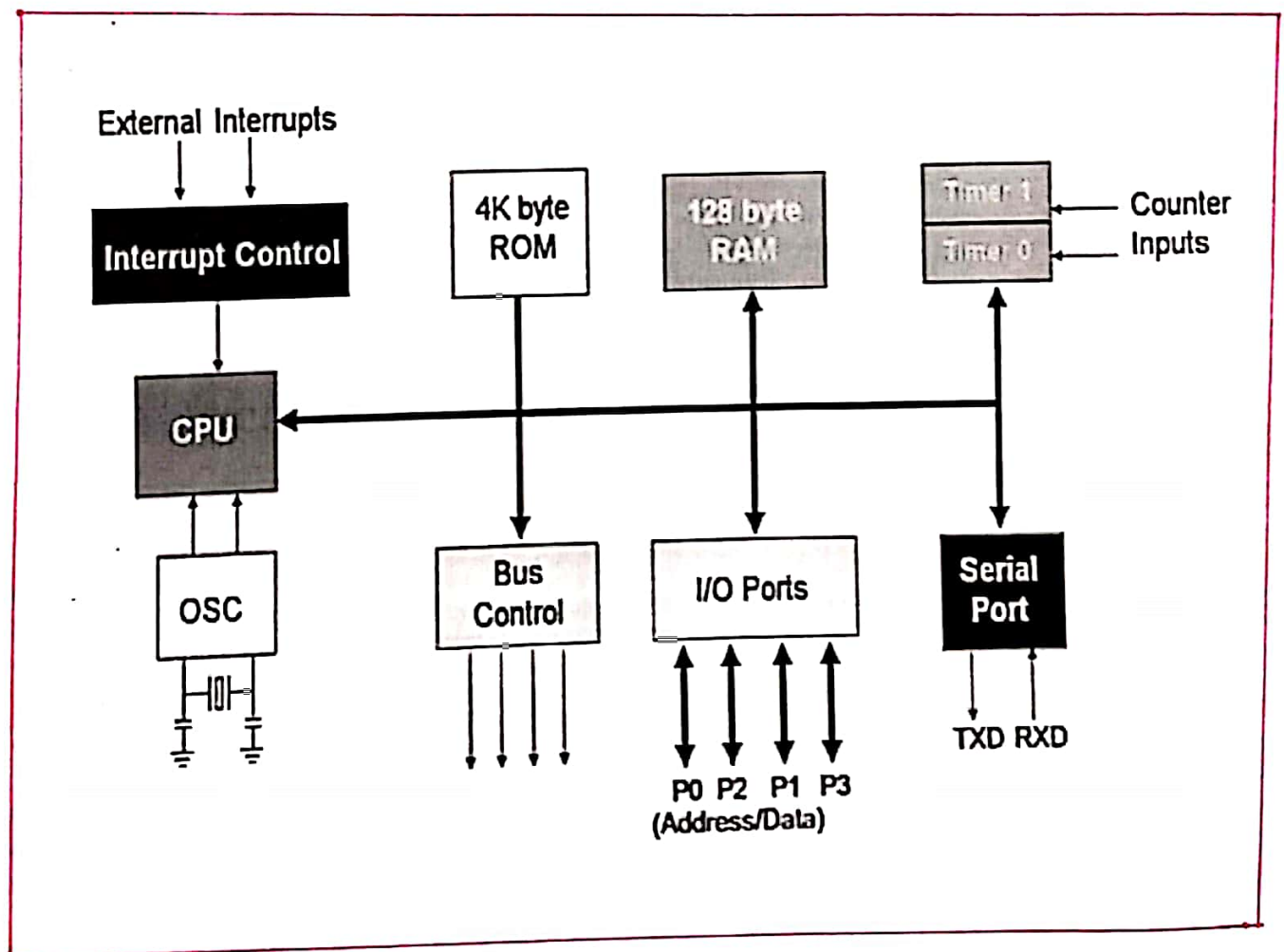
- 1> System requirement. i.e to understand what is to be achieved.
- 2> Application requirement. i.e input & output device requirement.
- 3> Electrical requirement.
- 4> Speed of operation.
- 5> Communication
- 6> operator interface
- 7> physical Environment.

What a Microcontroller is?

A microcontroller is an integrated chip designed under Very Large Scale Integration technique that consists of a processor with other peripheral units like memory, I/O port, timer, decoder, ADC etc. A microcontroller is basically designed in such a way that all the working peripherals are embedded in a single chip with the processor.

Any programmable device holds a processor, memory, I/O ports and timer within it. But a microcontroller contains all these components embedded in a single chip. This single-chip manages the overall operation of the device.

3. 8051 Microcontroller Architecture:



Basic components present internally inside 8051 Microcontroller architecture are:

CPU (Central Processing Unit): CPU act as a mind of any processing machine. It synchronizes and manages all processes that are carried out in microcontroller. User has no power to control the functioning of CPU. It interprets the program stored in ROM and carries out from storage and then performs its projected duty. CPU manage the different types of registers available in 8051 microcontroller.

Interrupts: Interrupts is a sub-routine call that given by the microcontroller when some other program with high priority is request for acquiring the system buses the n interrupts occur in current running program.

Interrupts provide a method to postpone or delay the current process, performs a sub-routine task and then restart the standard program again.

Types of interrupt in 8051 Microcontroller:

Let's see the five sources of interrupts in 8051 Microcontroller:

- Timer 0 overflow interrupt - TF0
- Timer 1 overflow interrupt - TF1
- External hardware interrupt - INT0
- External hardware interrupt - INT1
- Serial communication interrupt - RI/TI

Memory: For operation Micro-controller required a program. This program guides the microcontroller to perform the specific tasks. This program installed in microcontroller required some on chip memory for the storage of the program.

Microcontroller also required memory for storage of data and operands for the short duration. In microcontroller 8051 there is code or program memory of 4 KB that is it has 4 KB ROM and it also comprise of data memory (RAM) of 128 bytes.

Bus: Bus is a group of wires which uses as a communication canal or acts as means of data transfer. The different bus configuration includes 8, 16 or more cables. Therefore, a bus can bear 8 bits, 16 bits all together.

Types of buses in 8051 Microcontroller:

Let's see the two types of bus used in 8051 microcontroller:

- **Address Bus:** 8051 microcontrollers is consisting of 16 bit address bus. It is generally be used for transferring the data from Central Processing Unit to Memory.
- **Data bus:** 8051 microcontroller is consisting of 8 bits data bus. It is generally be used for transferring the data from one peripherals position to other peripherals.

Oscillator: As the microcontroller is digital circuit therefore it needs timer for their operation. To perform timer operation inside microcontroller it required externally connected or on-chip oscillator. Microcontroller is used inside an embedded system for managing the function of devices. Therefore, 8051 uses the two 16 bit counters and timers. For the operation of this timers and counters the oscillator is used inside microcontroller.

Input/ Output port: 8051 consists of 4 parallel ports of 8 bit each thereby providing 32 input-output pins. All the 4 ports function bidirectional i.e., either input or output according to the software control.

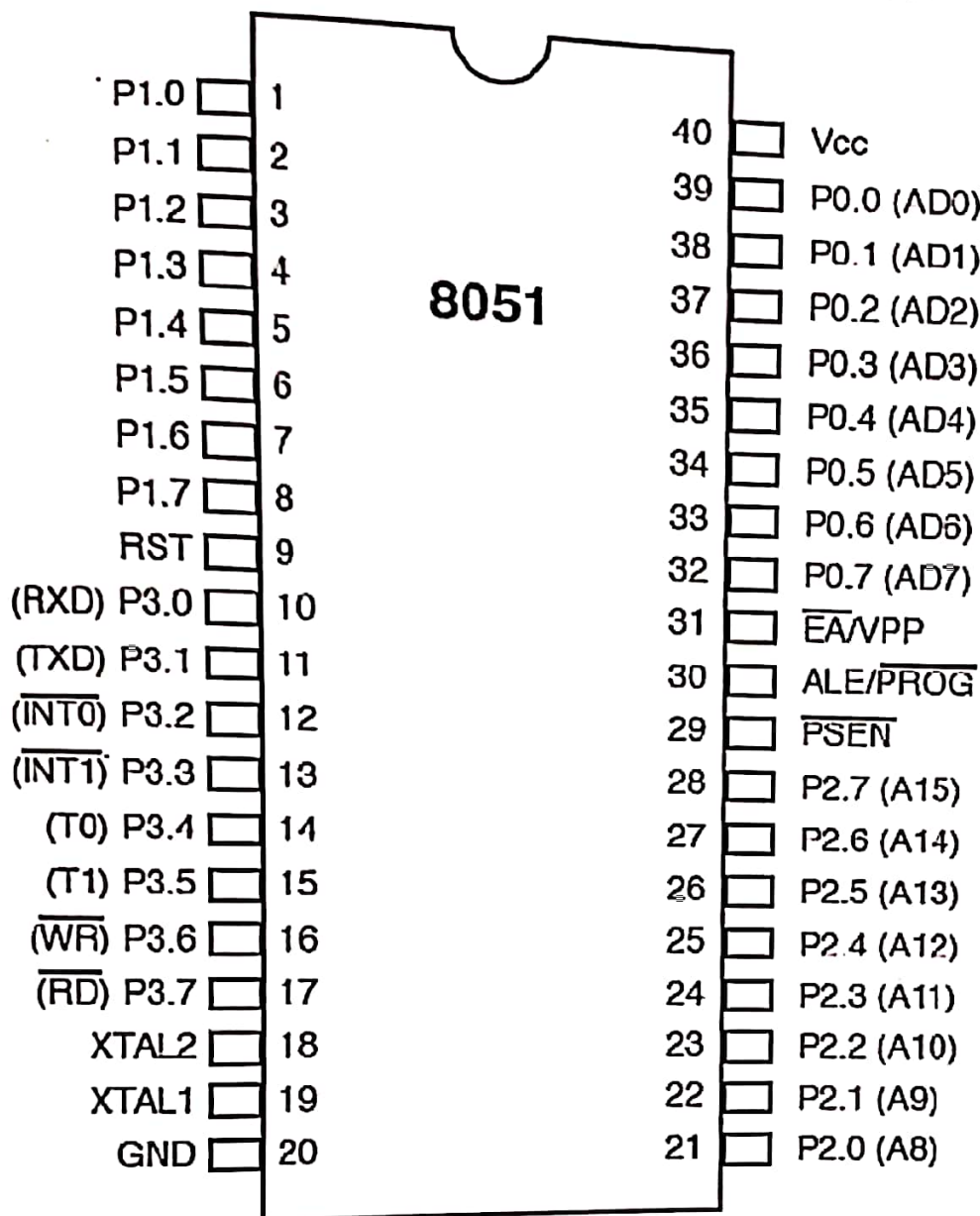
Timer and Control Unit: Timers are used to create a time gap or delay between 2 events. 8051 microcontroller consists of 2 timers of 16 bit each by which the system can produce two delays simultaneously in order to generate the appropriate delay. Basically, microcontrollers use **hardware delays** in which a physical device is used by the processor to produce the respective delay. And this physical device is known as a **timer**.

Serial port 8051: The 8051 contains UART – universal asynchronies receiver transmitter. • The serial port is full duplex. • It can transmit and receive simultaneously. • 2 port and 3 pins are used to provide the serial interface. P 3.0 is the receiver pin (RXD) P 3.1 is the transmitter pin (TXD)

Applications of Microcontroller 8051:

The use of **microcontroller** in various fields such as automobile, aeronautics, space, robotics, electronics, defense **application**, mobile communications, rail transport, industrial processing, and medical **applications** is rapidly increasing.

The pin diagram of 8051 microcontroller looks as follows –



- **Pins 1 to 8** – These pins are known as Port 1. This port doesn't serve any other functions. It is internally pulled up, bi-directional I/O port.
- **Pin 9** – It is a RESET pin, which is used to reset the microcontroller to its initial values.
- **Pins 10 to 17** – These pins are known as Port 3. This port serves some functions like interrupts, timer input, control signals, serial communication signals Rx/D and Tx/D, etc.
- **Pins 18 & 19** – These pins are used for interfacing an external crystal to get the system clock.
- **Pin 20** – This pin provides the power supply to the circuit.

- **Pins 21 to 28** – These pins are known as Port 2. It serves as I/O port. Higher order address bus signals are also multiplexed using this port.
- **Pin 29** – This is PSEN pin which stands for Program Store Enable. It is used to read a signal from the external program memory.
- **Pin 30** – This is EA pin which stands for External Access input. It is used to enable/disable the external memory interfacing.
- **Pin 31** – This is ALE pin which stands for Address Latch Enable. It is used to demultiplex the address-data signal of port.
- **Pins 32 to 39** – These pins are known as Port 0. It serves as I/O port. Lower order address and data bus signals are multiplexed using this port.
- **Pin 40** – This pin is used to provide power supply to the circuit.